SSH Key Setup

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An SSH key is a secure access credential used in the SSH protocol and establishes a secure and encrypted connection to Engaging. This page is for those who wish to implement SSH key authentication on top of general MIT Kerberos authentication.

SSH keys consist of a pair: a public key and a private key.

- **Public Key**: This key can be shared freely and is used to encrypt data that only the corresponding private key can decrypt.
- **Private Key**: This key must be kept secure and private. It is used to decrypt data encrypted with the corresponding public key and to prove the identity of the user during the SSH authentication process.

When you attempt to connect to an HPC system using SSH key authentication, the system uses your public key to initiate a challenge that can only be answered correctly using your private key. If the correct response is received, the system verifies your identity and grants access.

Using the key and lock analogy, the private key is like your key, and the public key is like a lock you might place on a gym locker, which would be like your account on Engaging. You can leave the lock locked on the locker at the gym, because no one can open the lock without the key, but you wouldn't want to share your key with anyone else, because then they could get into your locker.

Do Not Share Your Private Key

Your private key should never be shared with anyone. If someone else obtains your private key, they could potentially gain unauthorized access to any system your key is associated with.

Note

By default you will be prompted to enter your MIT Kerberos password even if you have set up SSH keys. Logging into ORCD OnDemand will allow you to log in with an ssh key and without a password for a short period of time.

Checking for Existing SSH Keys

Before you generate an SSH key, you should check for existing SSH keys.

1. Open your local terminal.
2. Run the following command to view all existing SSH keys:

```
ls -al ~/.ssh
```

3. If you see a list of files, you have existing SSH keys. If you receive an error that ~/.ssh doesn't exist, you do not have an existing SSH key pair in the default location. You can create a new SSH key pair in the next step.

Generating SSH Keys

If you do not have an existing SSH key, follow these steps.

1. Open your local Terminal.
2. Run the following command to generate an RSA key:

```
ssh-keygen -t rsa
```

3. **Save the key pair:** You will be prompted to enter a file path to save the key. Press `Enter` to accept the default location:

```
Enter a file in which to save the key (/home/your_username/.ssh/id_rsa):
```

4. **Passphrase:** You will be asked to enter a passphrase for additional security. You can either enter a passphrase or leave it empty and press `Enter`:

```
Enter passphrase (empty for no passphrase):
```

> **Note**
>
> Engaging, we recommend setting a passphrase as it provides extra security for your account by helping to prevent someone else from using your SSH keys. When you create a passphrase, your private key can only authenticate into your account when the correct passphrase is provided during login. **Since you set your passphrase on your system, ORCD staff cannot help you remember or reset the passphrase for your SSH keys. You must create new SSH keys if you cannot remember your passphrase.**

Uploading SSH Key on Engaging

To upload your SSH key on Engaging, you must update the `authorized_keys` file on Engaging via terminal. Alternatively, you have the option to use OnDemand.

If your system is a MAC or Linux machine, you can use the `ssh-copy-id` command to get your key onto the authorized_keys file on engaging. Windows users can try using git bash or WSL to access the `ssh-copy-id` command.

Terminal with ssh-copy-id

This method reduces the risk of typos in your authorized_keys file. It also correctly sets the permissions of the authorized_keys file in your authorized_keys file on engaging.

To add your SSH key with the Terminal and `ssh-copy-id`, please follow the steps outlined below (replace username with your Kerberos Login):

1. Open your terminal on your MAC or Linux machine (use git bash or WSL on Windows)
2. type

   ```
   ssh-copy-id [username]@orcd-login.mit.edu
   ```

   enter password, then select your DUO 2 Factor Authentication method.
3. If successful, you will get a message indicating that your key has been copied and you should now be able to use your ssh key.

Terminal manual copy

To add your SSH key via Terminal, please follow the steps outlined below:

1. Login to an HPC system login-node using MIT Kerberos Login.
2. On your local machine, copy the contents of your public key (`~/.ssh/id_rsa.pub`):

```
cat ~/.ssh/id_rsa.pub
```

> Make sure to copy the entire line starting with ssh-rsa and ending with your email address or comment.

3. On the head-node, append the copied contents to the authorized_keys file located at `/home/[username]/.ssh/authorized_keys` . You can use any text editor of your choice, such as nano and vim. For example, if you're using `nano` , the command would be:

```
nano /home/[username]/.ssh/authorized_keys
```

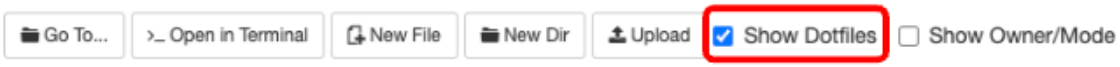
Note

Do not remove anything already present in the authorized_keys file. Be careful to append your key to the end of the file rather than replacing its contents.

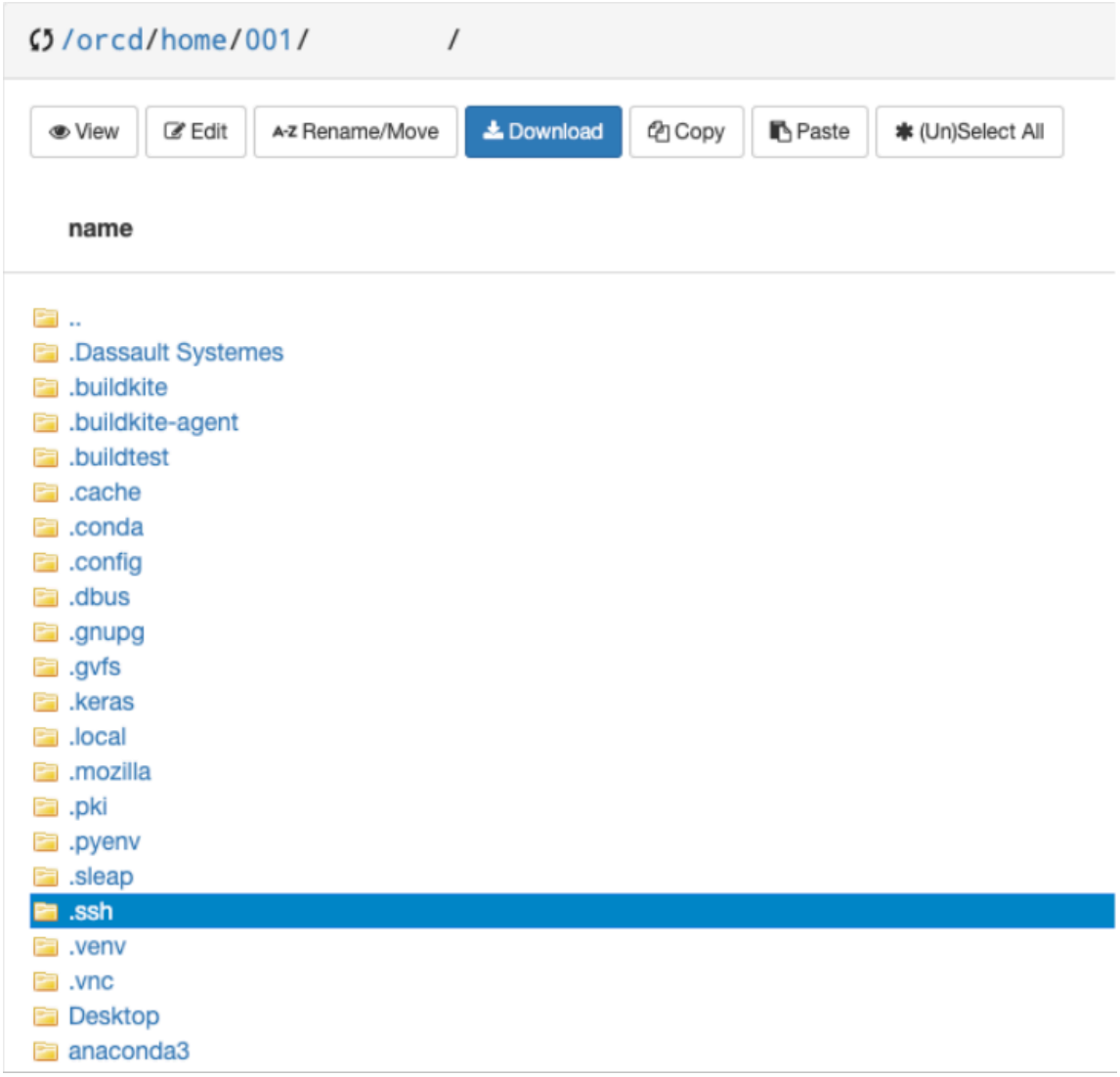
OnDemand

To add your SSH key to Engaging via OnDemand, please follow the steps outlined below:

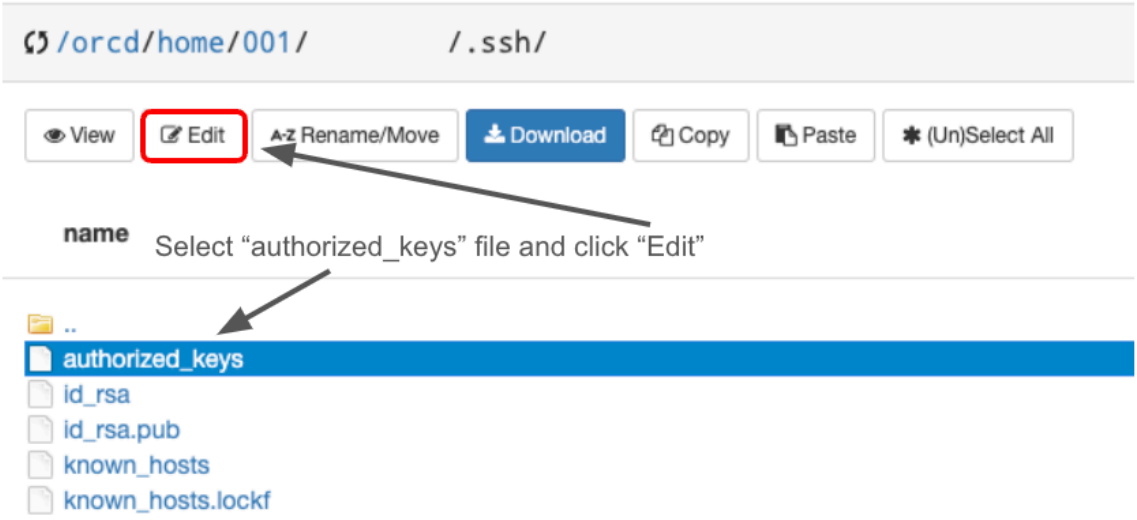
1. Login to ORCD OnDemand through the portal.
2. Once logged in, navigate to `Files` and `Home Directory` .
3. On the top right corner, check `Show Dotfiles` .



4. Click on the `.ssh` folder.



5. Locate and edit the `authorized_keys` file to add your new key.



Note

Do not remove anything already present in the authorized_keys file. Be careful to append your key to the end of the file rather than replacing its contents.

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